CHAPTER FOUR

DATA PRESENTATION AND DISCUSSION

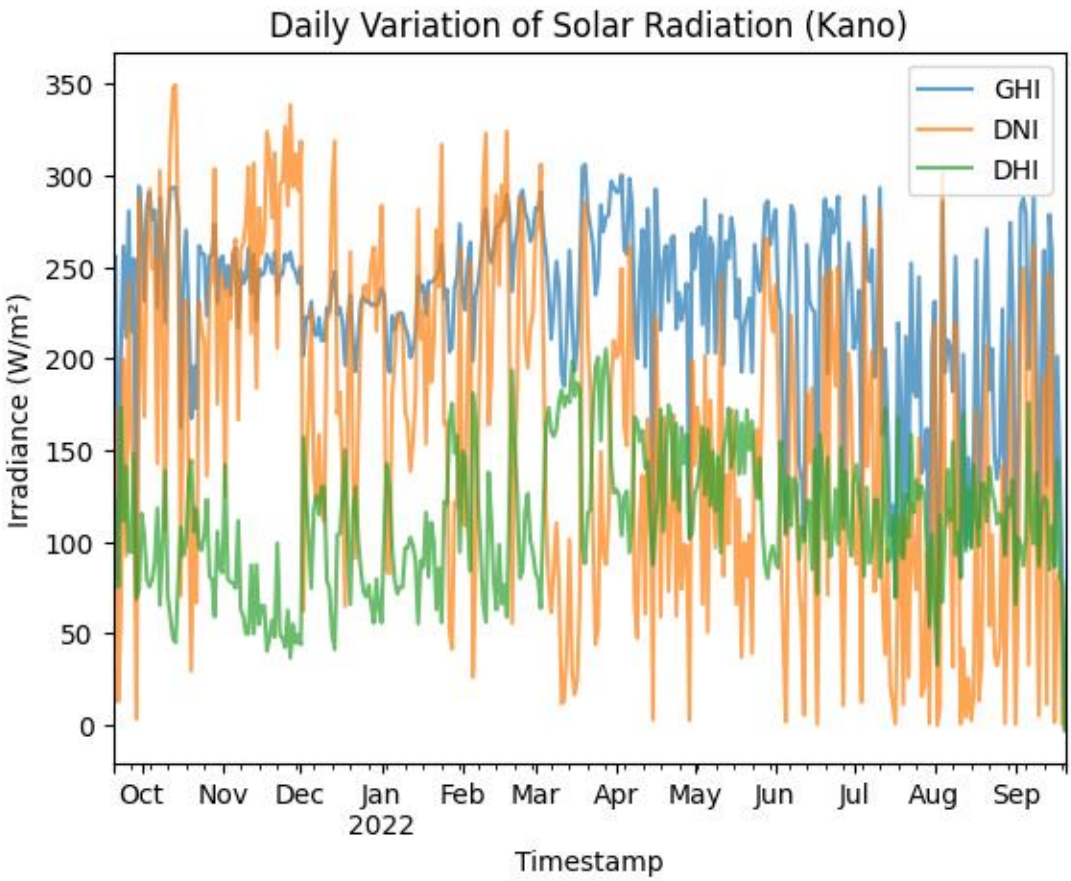
4.1 Introduction

This chapter presents the results of the statistical and graphical analyses conducted on the solar radiation dataset for Kano State, covering the period from September 20, 2020, to September 20, 2024. The analyses were guided by the study objectives, which focused on examining the daily, monthly, and seasonal variations of solar radiation (GHI, DNI, and DHI), evaluating the correlation among the radiation parameters, and comparing solar radiation values across wet and dry seasons. The findings are presented in tables and figures for clarity and are discussed in line with the research questions.

The results are interpreted not only in terms of statistical output but also in relation to their practical implications for solar energy development in Kano State. Emphasis is placed on identifying patterns of variability, examining the interrelationships between radiation parameters, and highlighting seasonal differences that affect solar resource availability. By linking the findings to existing literature, the discussion offers insights into how solar radiation in Kano aligns with broader studies on solar energy potential in Northern Nigeria.

4.2 Daily, Monthly, and Seasonal Variations of Solar Radiation in Kano State

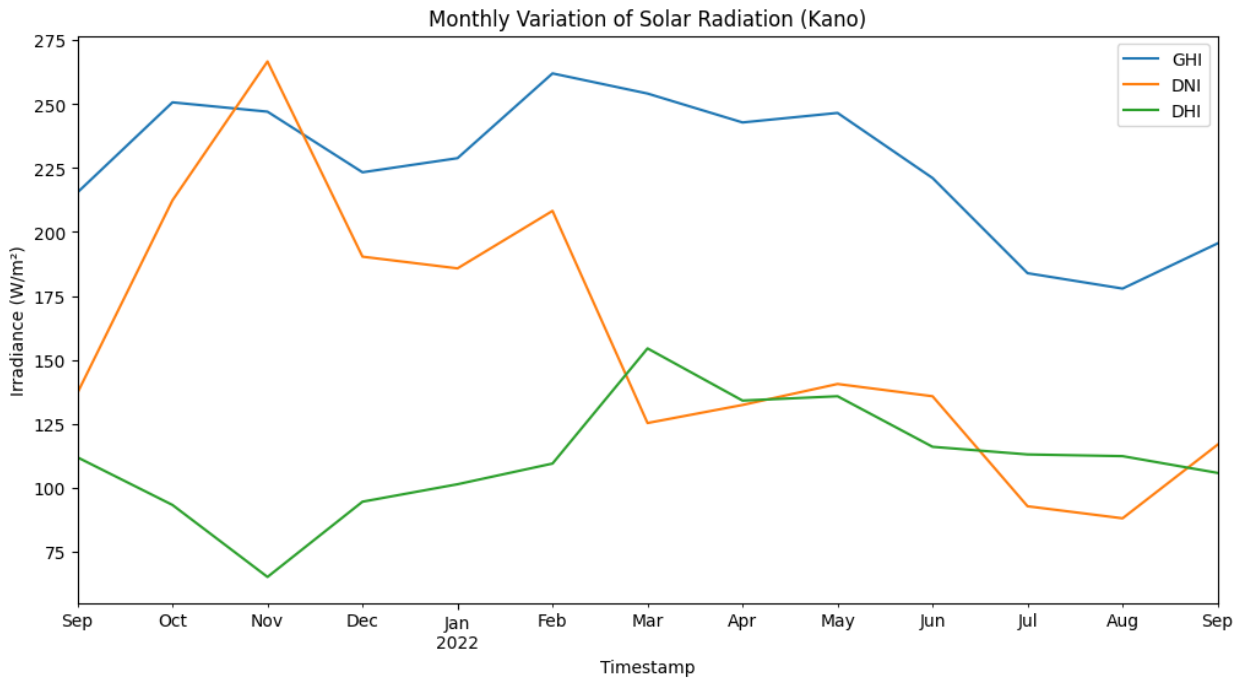
Fig 1: Daily Variation of Solar Radiation in Kano State, Nigeria



The daily variation of solar radiation in Kano is presented in Figure 4.1, which shows the fluctuations of Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), and Diffuse Horizontal Irradiance (DHI) over the study period (September 2021–September 2022). From the plot, GHI and DNI generally recorded higher intensities during the dry season months (November–April), reaching peaks above 300 W/m², while the wet season (May–October) exhibited lower and more fluctuating values, often below 150 W/m². This variation is explained by cloud cover and precipitation during the wet season, which reduces direct solar radiation and increases diffusion.

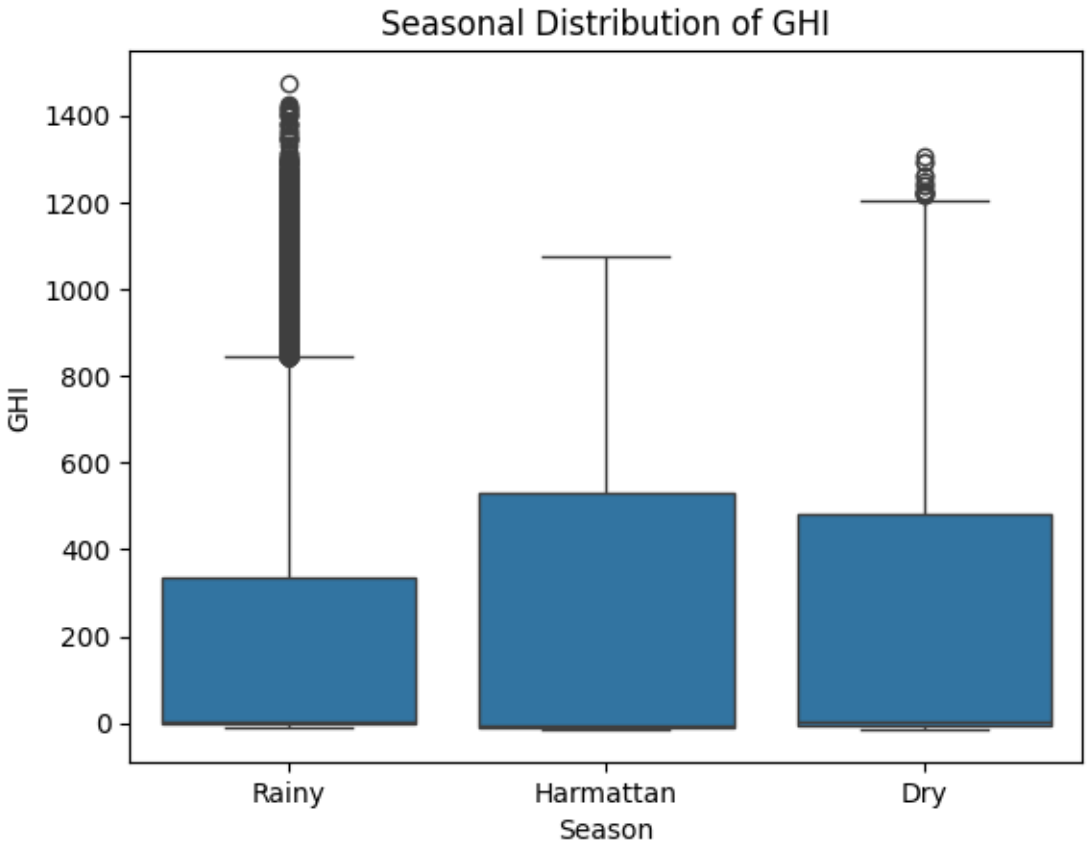
DHI showed a relatively smoother pattern compared to GHI and DNI, reflecting the influence of atmospheric scattering, especially during cloudy and rainy days.

Fig 2: Monthly Variation of Solar Radiation in Kano State, Nigeria



The monthly variation, displayed in Figure 4.2, further highlights these seasonal differences. For example, GHI and DNI peaked around November to February, consistent with clearer skies and reduced rainfall during the Harmattan and dry months. In contrast, during July and August, which correspond to peak rainfall, both GHI and DNI dropped significantly below 100 W/m², indicating low solar availability. DHI remained comparatively higher during these rainy months, suggesting that most of the solar energy reaching the surface was diffuse. These patterns underscore the influence of meteorological conditions on the temporal distribution of solar energy in Kano.

Fig 3: Seasonal Distribution of GHI in Kano State, Nigeria



The seasonal distribution of GHI, presented in Figure 4.3, provides a statistical summary of solar availability across the rainy season, Harmattan, and dry season. The boxplots show that the dry season and Harmattan had higher median GHI values (above 200 W/m²), with wider ranges indicating greater variability due to intermittent cloudiness and dust during Harmattan. Conversely, the rainy season recorded much lower medians (around 100 W/m²) with fewer extreme values, confirming that solar radiation availability is significantly reduced during wet periods.

Table 4.1: Summary Statistics of Daily Averages of Solar Radiation and Meteorological Variables (Sept 2021–Sept 2022)

Variable	Count	Mean	Std. Dev.	Min	25%	50%	75%	Max
GHI (W/m²)	366	227.69	133.48	0.00	109.21	229.67	309.05	905.73
DNI (W/m²)	366	157.75	90.77	0.00	83.55	172.64	217.25	931.78
DHI (W/m²)	366	111.30	79.74	0.00	38.55	101.97	173.29	397.81
Tamb (°C)	366	25.91	2.79	19.79	23.85	25.62	27.81	33.60
RH (%)	366	51.47	24.59	9.39	33.38	45.62	75.48	93.39
WS (m/s)	366	2.07	0.75	0.00	1.53	2.00	2.46	5.22

To complement the graphical results, the summary statistics of the key radiation and meteorological variables are shown in Table 4.1. GHI recorded a daily mean of 227.69 W/m², while DNI and DHI averaged 157.75 W/m² and 111.30 W/m² respectively. The maximum GHI observed was 905.73 W/m², while DNI peaked at 931.78 W/m² and DHI at 397.81 W/m², reflecting the high solar potential during clear-sky dry season days. Among the meteorological parameters, average ambient temperature (Tamb) was 25.91°C, relative humidity (RH) averaged 51.47%, and mean wind speed (WS) was 2.07 m/s, with gusts reaching higher extremes. These results affirm Kano's strong potential for solar energy exploitation, particularly during the dry months.

4.3 Correlation Analysis of Solar Radiation Parameters

To evaluate the interrelationships among the solar radiation parameters, correlation analysis was carried out on Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), and Diffuse Horizontal Irradiance (DHI).

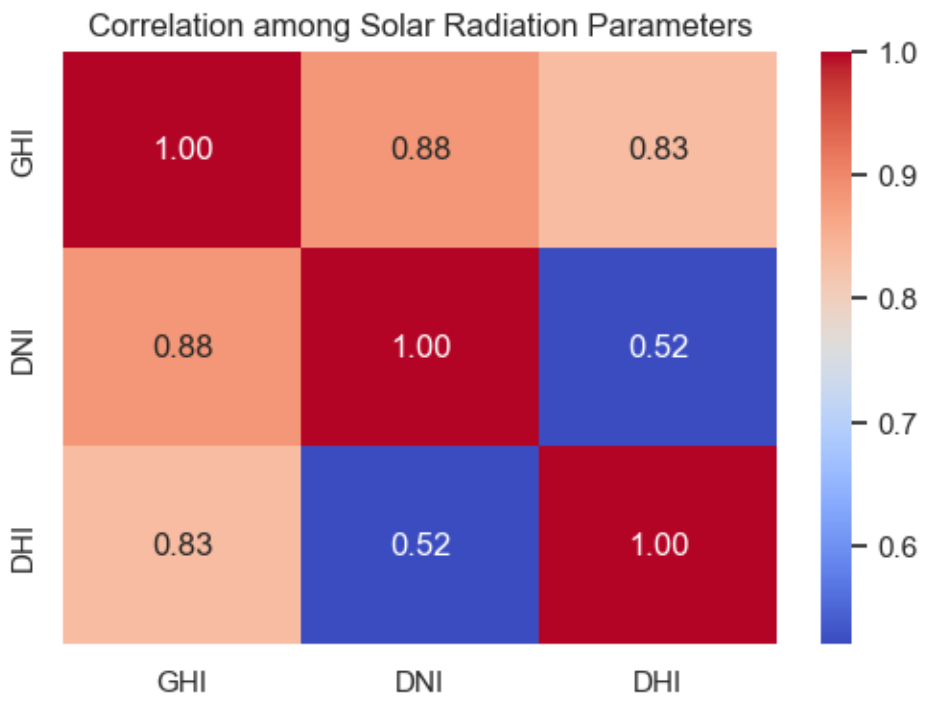
Table 4.2: Correlation Matrix of Solar Radiation Parameters

	GHI	DNI	DHI
GHI	1.00	0.88	0.83

DNI	0.88	1.00	0.52
DHI	0.83	0.52	1.00

The correlation matrix shown in Table 4.2 indicates that all three radiation parameters are positively correlated, although the strength of their relationships varies. GHI exhibited a strong positive correlation with DNI (r = 0.88) and with DHI (r = 0.83), suggesting that increases in global radiation are often accompanied by rises in both direct and diffuse components. DNI and DHI, however, showed a weaker correlation (r = 0.52), which is expected since high DNI (clear-sky conditions) typically coincides with lower DHI, while cloudy conditions increase DHI and reduce DNI.

Fig 4: Correlation among Solar Radiation Parameters in Kano State, Nigeria



The correlation heatmap in Figure 4.4 further visualizes these relationships, confirming that GHI is strongly influenced by both DNI and DHI. The pairwise scatterplots in Figure 4.5 provide

additional insight into the nature of these associations. The relationship between GHI and DNI is shown to be fairly linear, with most points clustering along an upward trend. Similarly, GHI and DHI display a strong upward pattern, though more dispersed compared to DNI. Conversely, the DNI–DHI scatterplot reflects a less consistent trend, showing greater spread and weaker alignment, in line with the moderate correlation value obtained.

Fig 5: Pairwise Relationships of GHI, DNI, and DHI in Kano State, Nigeria

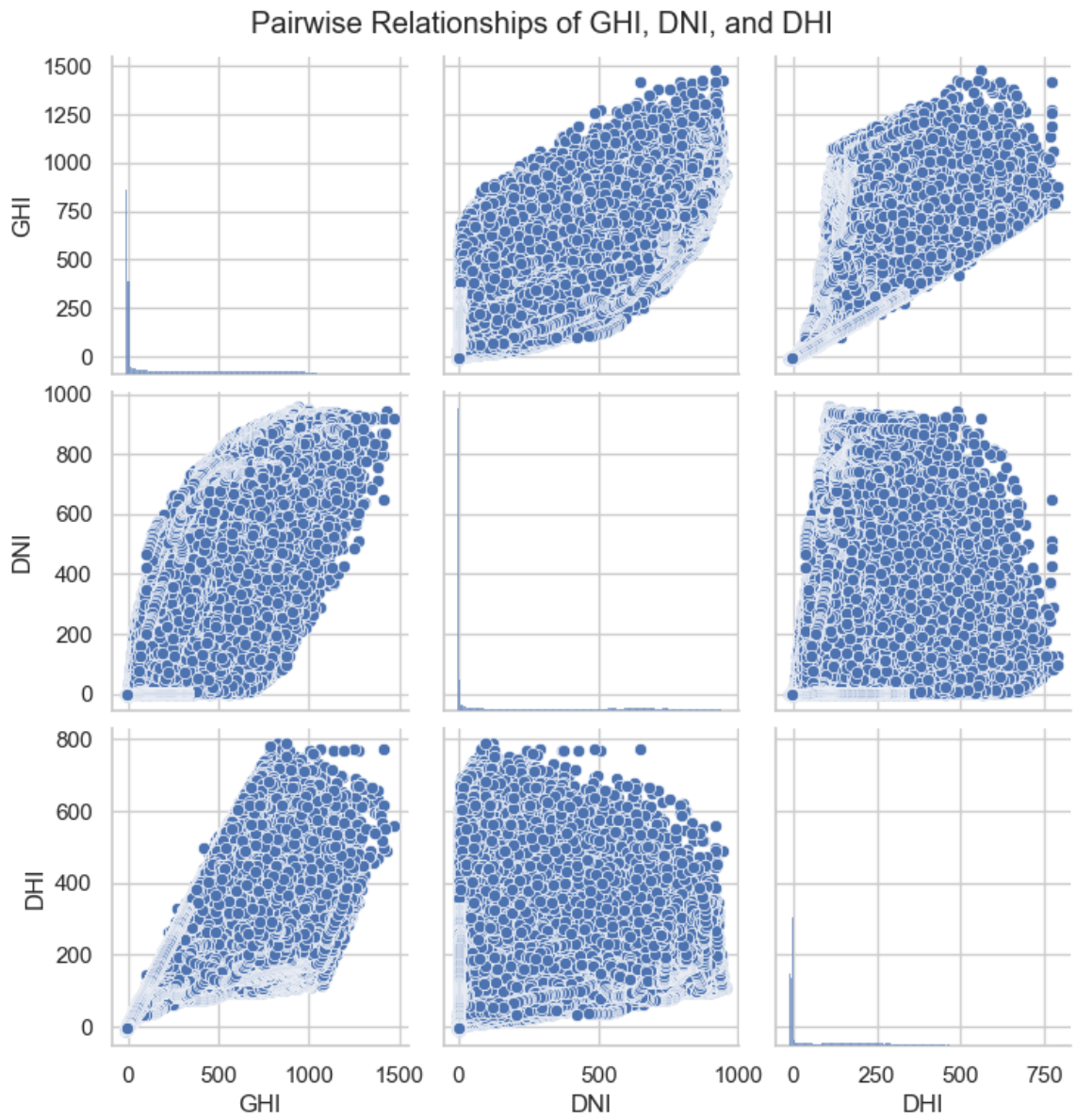
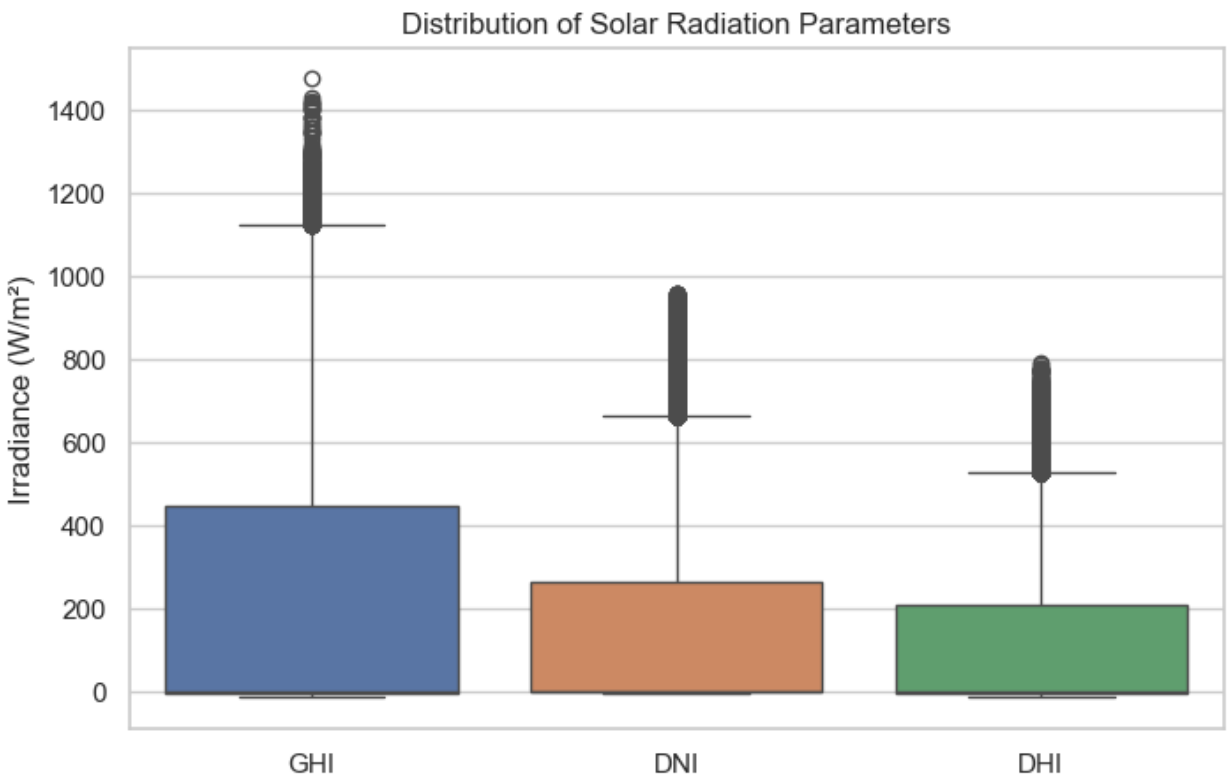


Fig 6: Distribution of Solar Radiation Parameters in Kano State, Nigeria



The distribution of the three parameters is illustrated in Figure 4.6 using boxplots. GHI displays the highest median and widest range of values, confirming its role as the most significant measure of total solar energy availability. DNI also shows a wide spread, with several high-value outliers corresponding to clear-sky conditions during dry months. DHI, while lower on average, exhibits variability during cloudy and rainy periods, with many moderate outliers observed. These results highlight the complementary nature of direct and diffuse radiation in contributing to global irradiance.

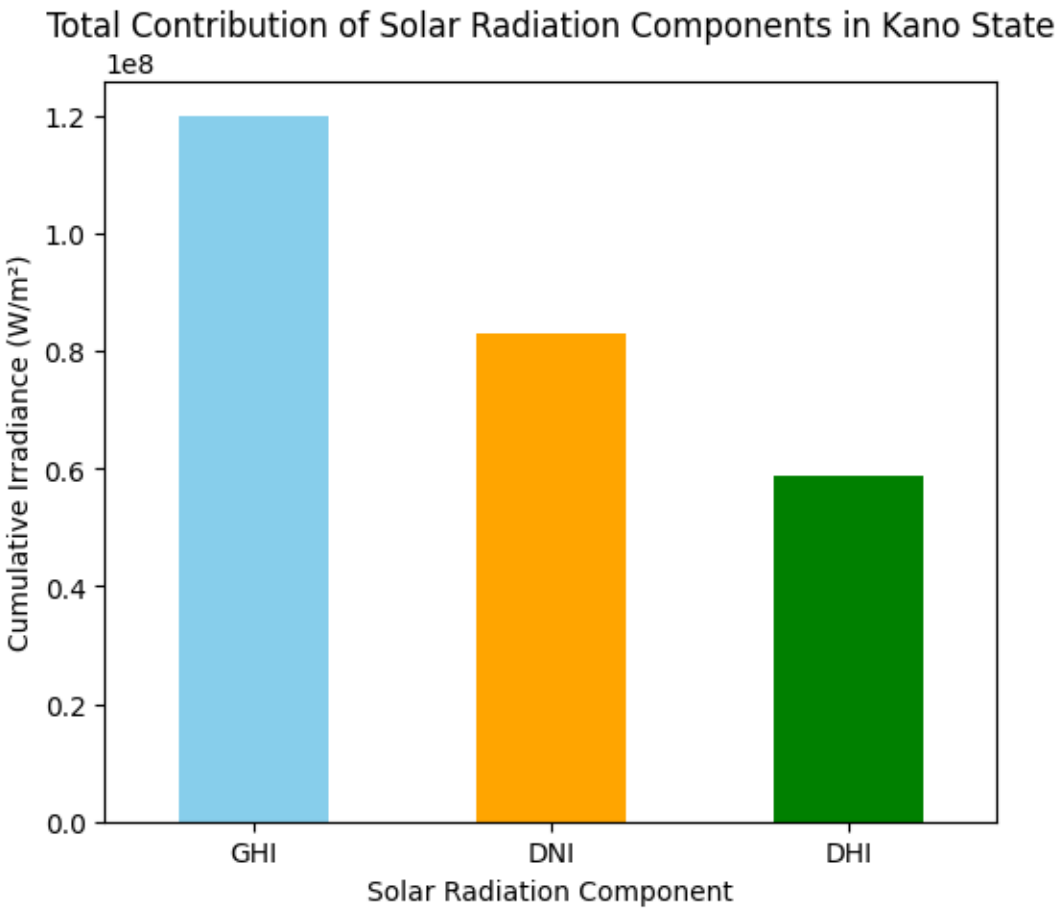
Overall, the correlation analysis emphasizes that GHI in Kano is jointly influenced by both DNI and DHI, with DNI contributing more strongly under clear-sky conditions, while DHI becomes

more important during cloudy or rainy days. This reinforces the importance of considering both components in solar resource assessments.

4.4 Contribution of Solar Radiation Components

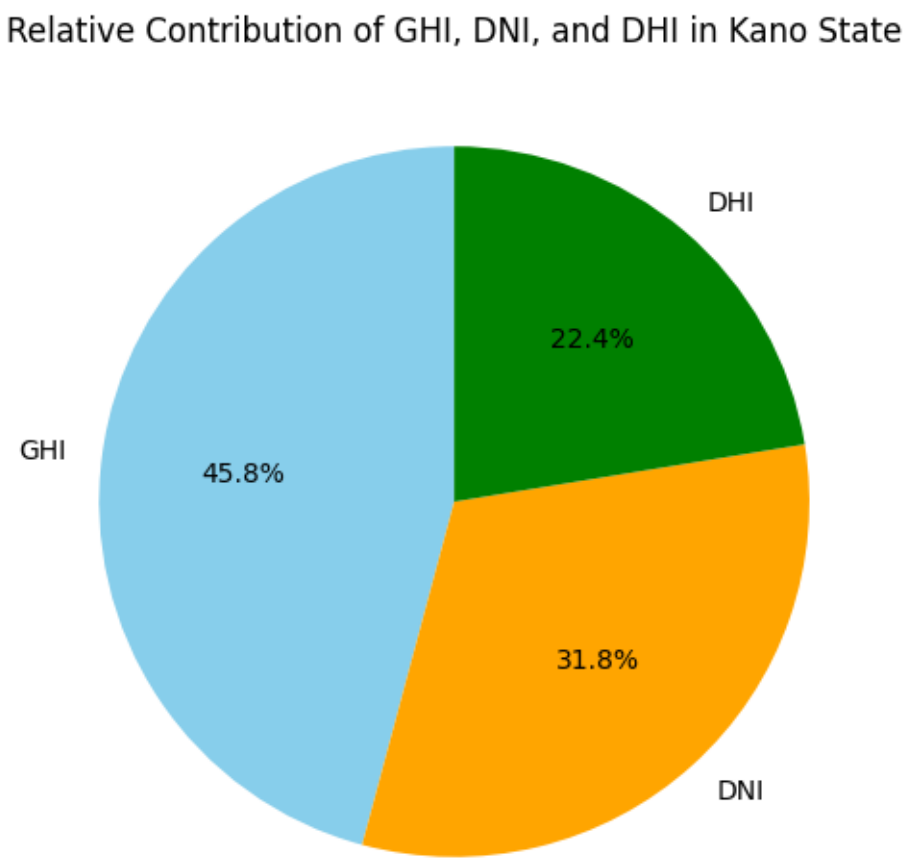
To evaluate the overall contribution of the solar radiation components in Kano, the cumulative and seasonal averages of Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), and Diffuse Horizontal Irradiance (DHI) were analyzed. The results presented in Figures 4.7 to 4.9 provide insights into how these three components contribute individually and collectively to solar energy potential in the study area.

Fig 7: Total Contribution of Solar Radiation Components in Kano State, Nigeria



As shown in Figure 4.7, the cumulative contribution highlights that GHI is the dominant parameter, recording the highest total irradiance of approximately 1.2×10^6 Wh/m². This is followed by DNI with about 0.8×10^6 Wh/m², while DHI contributes the least at roughly 0.6×10^6 Wh/m². This distribution demonstrates that while all three parameters are important, GHI provides the most comprehensive measure of solar availability, as it accounts for both direct and diffuse components.

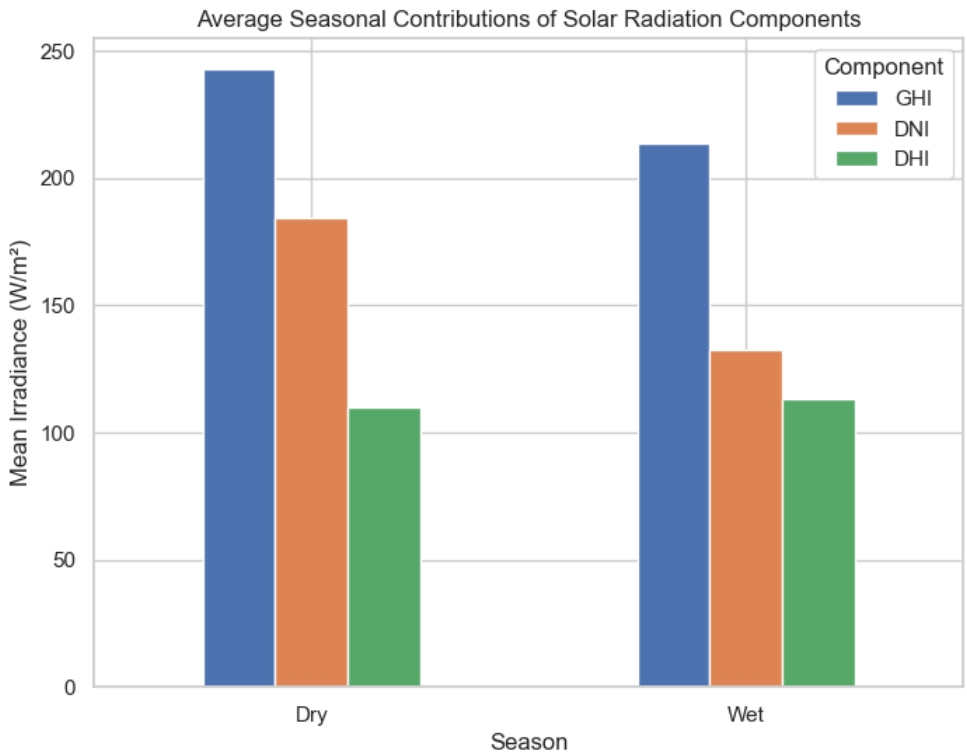
Fig 8: Relative Contribution of GHI, DNI, and DHI in Kano State, Nigeria



The relative contributions of the parameters are illustrated in Figure 4.8 using a pie chart. GHI contributes 45.8% of the total solar energy, followed by DNI at 31.8%, and DHI at 22.4%. This breakdown shows that nearly half of the solar radiation in Bauchi is global, with DNI and DHI

complementing it depending on sky conditions. DNI's substantial share indicates that Kano experiences frequent clear-sky periods, making direct radiation a major contributor. At the same time, the notable fraction of DHI underscores the role of diffuse radiation during cloudy or rainy days.

Fig 9: Average Seasonal Contributions of Solar Radiation Components in Kano State, Nigeria



The seasonal comparison in Figure 4.9 provides further clarity on how contributions vary between the dry and wet seasons. During the dry season, GHI and DNI values are significantly higher, averaging about 230 W/m² and 180 W/m² respectively, compared to the wet season averages of 210 W/m² for GHI and 140 W/m² for DNI. DHI, on the other hand, is slightly higher in the wet season (110 W/m²) compared to the dry season (100 W/m²), reflecting the expected increase in diffuse radiation during periods of cloud cover and rainfall.

In summary, the findings from this objective indicate that while GHI remains the most dominant contributor to solar energy in Kano across all seasons, DNI is particularly influential during the dry season, and DHI plays a critical role during the wet season. These dynamics highlight the importance of considering both direct and diffuse components in the design and deployment of solar technologies in the region.

4.5 Discussion of Findings

The findings from the analysis of daily, monthly, and seasonal variations of solar radiation (Objective One) revealed significant fluctuations across different timescales in Kano. Daily and monthly patterns showed that GHI, DNI, and DHI values peaked during the dry season months, particularly between November and March, while lower values were observed during the wet season months, notably June through August. This result supports the assertion of Bala et al. (2019), who found that Northern Nigeria benefits from clear-sky conditions during the dry season, making it an attractive region for solar energy exploitation. Similarly, Mustapha and Umar (2021) observed that Kano exhibits one of the highest annual solar energy potentials in Nigeria, which is consistent with the high irradiance values recorded in this study.

In examining the relationships among solar radiation parameters (Objective Two), the correlation analysis indicated strong positive associations between GHI, DNI, and DHI, with GHI and DNI exhibiting the highest correlation coefficient (0.88). This suggests that periods of strong direct radiation also correspond to high global irradiance, while diffuse radiation contributes moderately depending on atmospheric conditions. These results align with the findings of Okonkwo et al. (2020), who noted that global radiation in Nigeria can be largely predicted from direct radiation under clear-sky conditions. Furthermore, the distribution patterns observed in this study

corroborate those of Nwokolo and Udo (2017), who highlighted the combined role of direct and diffuse components in accurately estimating total solar availability for photovoltaic applications.

The contribution analysis of GHI, DNI, and DHI (Objective Three) further established GHI as the most dominant parameter, accounting for 45.8% of total solar irradiance, followed by DNI (31.8%) and DHI (22.4%). These proportions demonstrate the crucial role of direct radiation in Kano's energy mix, particularly during dry months when clear skies prevail. This observation aligns with the results of Falayi and Rabiu (2019), who emphasized the predominance of DNI in Northern Nigeria due to its dry climate. At the same time, the relatively high share of DHI during the wet season reflects the findings of Adaramola (2016), who showed that diffuse radiation is critical for maintaining solar energy supply under cloudy conditions, ensuring more stable outputs for solar collectors and photovoltaic systems.

The seasonal analysis also revealed that solar energy potential in Kano is consistently higher during the dry season than in the wet season, with average GHI and DNI values significantly reduced during rainy periods. This agrees with the conclusions of Adekunle et al. (2020), who stressed that seasonal rainfall and high humidity in Nigeria reduce DNI levels, increasing the dominance of diffuse radiation. Importantly, the findings of this study provide additional evidence that seasonal dynamics must be carefully integrated into solar project planning, particularly in selecting appropriate solar technologies. For instance, photovoltaic systems may perform efficiently in both wet and dry conditions due to their ability to harness diffuse radiation, while concentrating solar systems dependent on DNI would be most effective during dry months.

Overall, the results of this study are consistent with the broader body of literature on solar energy in Nigeria, confirming that Kano has a high potential for solar exploitation due to its abundant radiation levels and favorable dry season conditions. However, the seasonal variability underscores

the need for adaptive energy strategies that can leverage both direct and diffuse radiation to ensure year-round efficiency. This study therefore contributes to bridging the gap in region-specific solar energy data for Kano, providing empirical evidence that supports earlier works while adding localized detail for improved planning and sustainable energy development.